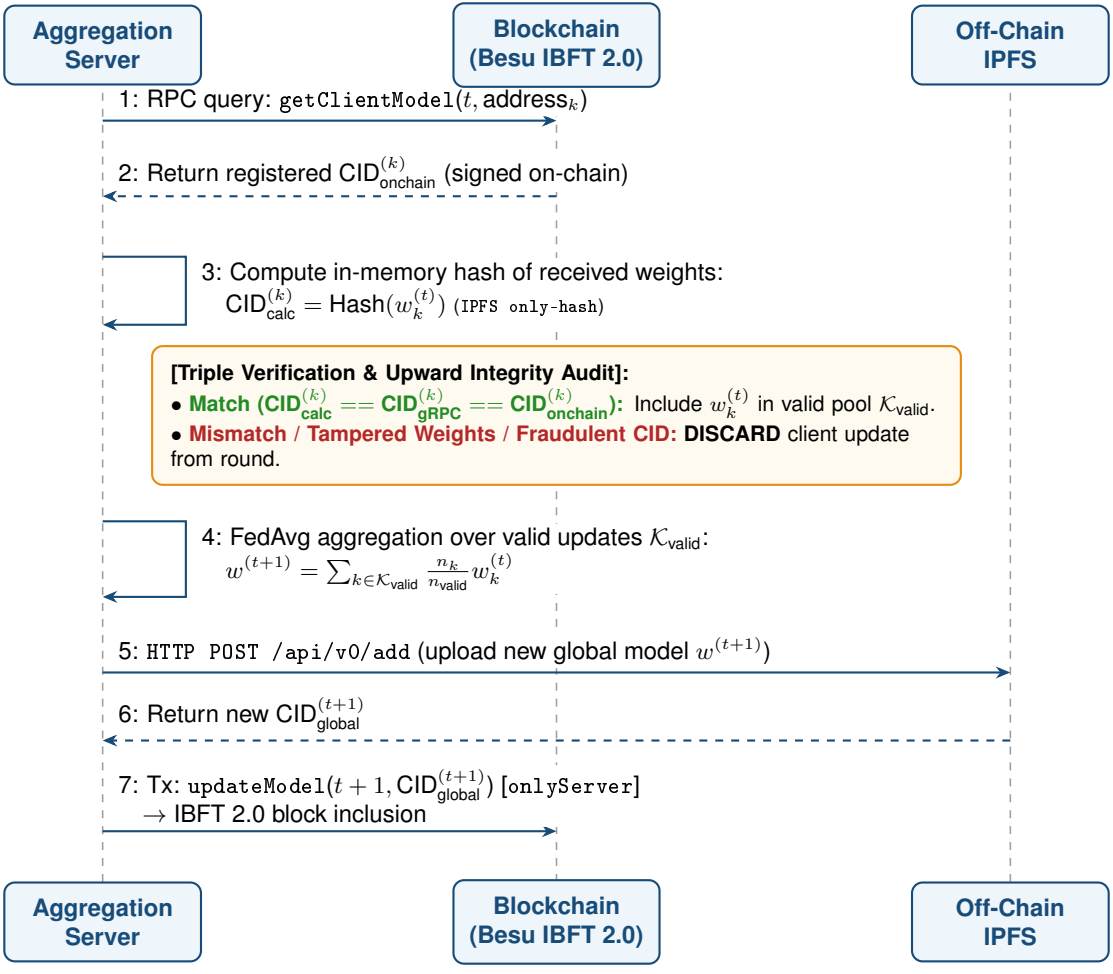

Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS
1: RPC query: getClientModel(t, address$_k$)
2: Return registered CID$^{(k)}_{\text{onchain}}$ (signed on-chain)
3: Compute in-memory hash of received weights:
CID$^{(k)}_{\text{calc}}$ = Hash($w_k^{(t)}$) (IPFS only-hash)
[Triple Verification & Upward Integrity Audit]:
• Match (CID$^{(k)}_{\text{calc}}$ == CID$^{(k)}_{\text{gRPC}}$ == CID$^{(k)}_{\text{onchain}}$): Include $w_k^{(t)}$ in valid pool $\mathcal{K}_{\text{valid}}$.
• Mismatch / Tampered Weights / Fraudulent CID: DISCARD client update from round.
4: FedAvg aggregation over valid updates $\mathcal{K}_{\text{valid}}$:
$w^{(t+1)} = \sum_{k \in \mathcal{K}_{\text{valid}}} \frac{n_k}{n_{\text{valid}}} w_k^{(t)}$
5: HTTP POST /api/v0/add (upload new global model $w^{(t+1)}$)
6: Return new CID$^{(t+1)}_{\text{global}}$
7: Tx: updateModel($t+1$, CID$^{(t+1)}_{\text{global}}$) [onlyServer]
→ IBFT 2.0 block inclusion
Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS